

Tejas Shivakumar Kasetty

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Profile

Student researcher at Université de Montréal / Mila - Québec AI Institute studying information-theoretic and cognitive principles underlying learning, reasoning, and compositional generalization.

Education

2022–Present	M.Sc. in Computer Science <i>Université de Montréal / Mila - Québec AI Institute</i> Co-supervised by Guillaume Lajoie and Dhanya Sridhar. GPA: 4.15/4.3.	Montreal, Canada
2014–2018	B.Tech. in Computer Science <i>PES University</i> Undergraduate thesis and research internship in computer vision at the Center for Pattern Recognition and Machine Intelligence (CPRMI).	Bengaluru, India

Publications

2026	Simplifying the Modeling of Arbitrary Conditionals in Natural Language Yinhan Lu*, Eric Elmoznino*, Léo Gagnon, Sarthak Mittal, Tejas Kasetty , Guillaume Lajoie <i>arXiv preprint; introduces AC-GPT for arbitrary conditional language modeling</i>
2026	A Compression Perspective on Simplicity Bias Tom Marty, Eric Elmoznino, Leo Gagnon, Tejas Kasetty , Mizu Nishikawa-Toomey, Sarthak Mittal, Guillaume Lajoie, Dhanya Sridhar <i>arXiv preprint</i>
2025	In-context Learning and Occam's Razor Eric Elmoznino*, Tom Marty*, Tejas Kasetty , Leo Gagnon, Sarthak Mittal, Mahan Fathi, Dhanya Sridhar, Guillaume Lajoie <i>International Conference on Machine Learning (ICML), poster</i>
2025	Next-Token Prediction Should be Ambiguity-Sensitive: A Meta-Learning Perspective Leo Gagnon, Eric Elmoznino, Sarthak Mittal, Tom Marty, Tejas Kasetty , Dhanya Sridhar, Guillaume Lajoie <i>ICML Workshop on Efficient Systems for Foundation Models; arXiv preprint</i>
2024	Evaluating Interventional Reasoning Capabilities of Large Language Models Tejas Kasetty* , Divyat Mahajan, Gintare Karolina Dziugaite, Alexandre Drouin, Dhanya Sridhar <i>NeurIPS CALM Workshop, poster</i>

Research Experience

Sept 2024–Present	Graduate Student Researcher <i>Mila - Quebec AI Institute</i> - Researching compositionality and compositional generalization in neural learning systems as part of Guillaume Lajoie's lab. - Developing data-ordering and curriculum-learning methods for better representations, using autoregressive/generative models to evaluate predictive efficiency. - Building PyTorch pipelines for policy learning over image/patch orders and downstream representation evaluation.	Montreal, Canada
July 2023–Aug 2024	Research Intern <i>Mila - Quebec AI Institute</i> - Evaluated causal and interventional reasoning abilities of large language models under controlled benchmark settings, separating memorization from reasoning. - Contributed to theory and experiments connecting in-context learning as meta-learning to Occam-style simplicity bias.	Montreal, Canada
May 2017–April 2018	Research Intern <i>Center for Pattern Recognition and Machine Intelligence (CPRMI) Lab, PES University</i> - Worked on computer vision research for face recognition and image deblurring during a summer internship and undergraduate thesis. - Implemented neural-network and image-processing methods for degraded visual data and restoration-quality evaluation.	Bengaluru, India

Industry Experience

May 2023–Nov 2023	Machine Learning Developer Intern <i>Intellisports Inc.</i> - Built sequence-modeling approaches for human activity recognition from inertial measurement unit (IMU) time-series data. - Developed models and evaluation pipelines for activity recognition and repetition counting in sports and fitness settings.	Montreal, Canada
June 2018–Jan 2020	Software Development Engineer - Big Data and ML <i>MiQ</i> - Implemented Spark-based libraries and services supporting scalable ML workflows on large-scale advertising and analytics data. - Helped analysts and data scientists automate data pipelines and train machine-learning models in distributed computing environments.	Bengaluru, India
Jan 2018–June 2018	Software Development Intern - Full Stack <i>MiQ</i> - Developed a web application for creating, running, and scheduling workflows. - Refactored backend services from a monolithic architecture to microservices and migrated deployment to Kubernetes.	Bengaluru, India

Selected Projects

2023–Present	Quick-Recipe: Knowledge Extraction from Cooking Videos <i>Université de Montréal</i> - Explored multimodal text-video models for extracting objects, actions, and procedural knowledge from cooking videos. - Used temporal segment models and attention mechanisms for text-to-video temporal grounding.	Montreal, Canada
2022	NHL Analytics <i>Université de Montréal</i> - Built an end-to-end pipeline for NHL play-by-play data from 2018–2021, including querying, visualization, and live-game analytics. - Trained an ensemble model for shot/miss/goal prediction and deployed a web application for real-time game insights.	Montreal, Canada

Technical Skills

Languages	Python, Scala, Java, C, C++, MATLAB, LaTeX
ML / Research	PyTorch, PyTorch Lightning, Hydra, TensorFlow, scikit-learn, NumPy, Pandas, Transformers, autoregressive models, reinforcement learning, causal inference, representation learning
Systems / Tools	Spark, Git, Jupyter, Databricks, Qubole Notebooks, Kubernetes

Awards, Service, and Leadership

2019	Best Newcomer Award , MiQ, Bengaluru, India.
2023–2024	Student Affairs Committee Member, Content Manager , UNIQUE, Montreal. Supported student programming at the intersection of neuroscience and AI in Quebec.
2023	Organizer / Volunteer , UNIQUE Student Symposium. Assisted with event logistics for a student research symposium in neuroscience and AI.
2014–2017	PES University Football Team . Represented the university in multiple tournaments.